

MATH

Area of quadrilateral ABCD
 $= \frac{1}{2} \times (\text{diagonal}) \times (\text{sum of its offsets})$

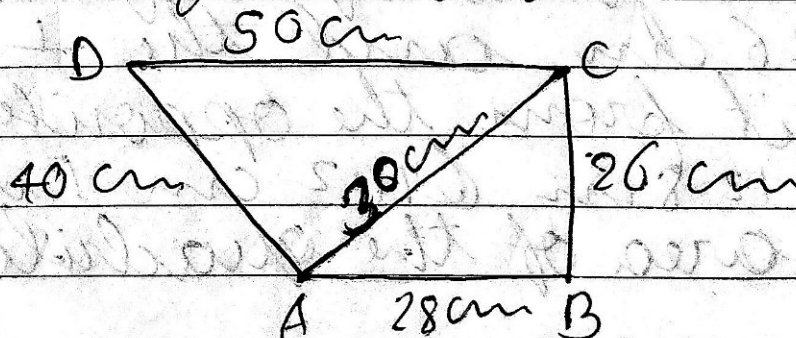
$$= \frac{1}{2} \times 26 \times (11.2 + 12.8) \text{ cm}^2$$

$$= \frac{1}{2} \times 26 \times 24 \text{ cm}^2$$

$$= \boxed{312 \text{ cm}^2}$$

18/12/23

- 10) In a quadrilateral ABCD, $AB = 28 \text{ cm}$, $BC = 26 \text{ cm}$, $CD = 50 \text{ cm}$, $DA = 40 \text{ cm}$ and diagonal $AC = 30 \text{ cm}$. Find the area of the quadrilateral.



for $\triangle ABC$;

$$S = \frac{1}{2} (28 + 26 + 30) \text{ cm}$$

$$= \frac{1}{2} \times 84 \text{ cm} = 42 \text{ cm}$$

area of $\triangle ABC$;

$$= \sqrt{42(42-28)(42-26)(42-30)} \text{ cm}^2$$

$$= \sqrt{42 \times 14 \times 16 \times 12}$$

$$= \sqrt{14 \times 3 \times 14 \times 16 \times 4 \times 3} \text{ cm}^2$$

$$= 14 \times 3 \times 4 \times 2 \text{ cm}^2$$

$$= \boxed{336 \text{ cm}^2}$$

for $\triangle ACD$;

$$S = \frac{1}{2} (30 + 50 + 40) \text{ cm}$$

$$= \frac{1}{2} \times 120 \text{ cm} = 60 \text{ cm}$$

area of $\triangle ABC$;

$$= \sqrt{60(60-30)(60-50)(60-40)} \text{ cm}^2$$

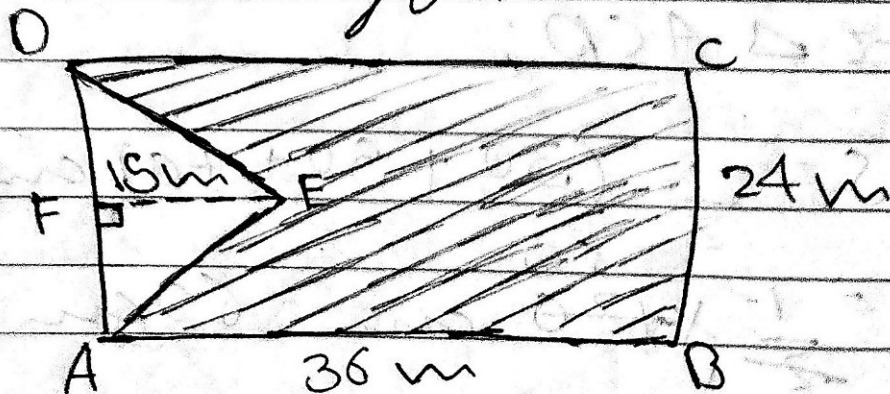
$$= \sqrt{60 \times 30 \times 10 \times 20} \text{ cm}^2$$

$$= \sqrt{20 \times 3 \times 3 \times 10 \times 10 \times 20} \text{ cm}^2$$

$$= 20 \times 3 \times 10 = \boxed{600 \text{ cm}^2}$$

area of the quadrilateral ABCD
 $= \text{area } (\triangle ABC + \triangle ACD)$
 $= (336 + 600) \text{ cm}^2$
 $= \boxed{936 \text{ cm}^2}$

- 21) In the given figure, ABCD is a rect. with length = 36m and breadth = 24m. In $\triangle ADE$, $EF \perp AD$ and $EF = 15\text{m}$. Calculate the area of the shaded region.



Area of Rect. ABCD
 $= 36 \times 24 \text{ m}^2$
 $= 864 \text{ m}^2$

Area of $\triangle AOE$;

$$= \frac{1}{2} \times 24 \times 15 \text{ m}^2$$

$$= 180 \text{ m}^2$$

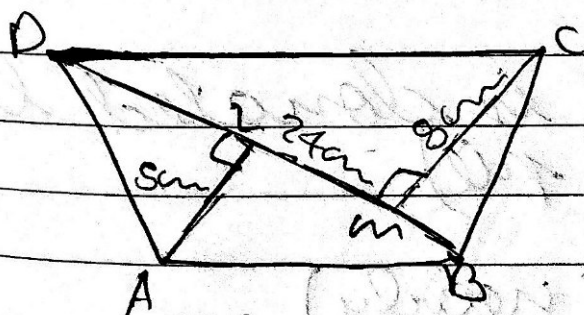
Area of the shaded region

$$= (\text{area of the rect.}) - (\text{area of } \triangle)$$

$$= (864 - 180) \text{ m}^2$$

$$= 684 \text{ m}^2$$

24) Find the area of a quadrilateral ABCD in which diagonal BD = 24 cm. $AL \perp BD$ and $CM \perp BD$ such that $AL = 5 \text{ cm}$ and $CM = 8 \text{ cm}$.



Area of the quad.

$$= \frac{1}{2} \times (\text{diagonal}) \times (\text{sum of its offsets})$$

$$= \frac{1}{2} \times 24 \times (5 + 8) \text{ cm}^2 = \frac{1}{2} \times 24 \times 13 \text{ cm}^2$$

$$= \underline{156 \text{ cm}^2}$$

Circumference and area of a circle.

Circumference: The perimeter of a circle is called its circumference.

Notes...

The ratio of the circumference of a circle and its ~~upper~~ diameter is always constant.

$$\frac{\text{Circumference}}{\text{Diameter}} = 3.14 \text{ (approx)}$$

The ratio is denoted by π (called pi)

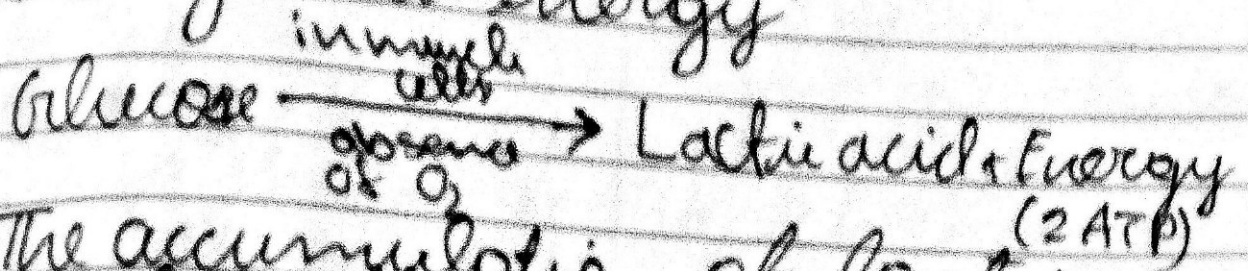
$$\begin{aligned} \pi &= 3.14 \text{ (nearly)} \\ &= \frac{22}{7} \text{ (nearly)} \end{aligned}$$

$$\therefore \frac{C}{2r} = \pi$$

$$\therefore C = 2\pi r$$

BIO.

along with energy.



- The accumulation of lactic acid in the muscle cells may result in muscle cramps.

5/12/21

Difference between the aerobic and anaerobic respiration

Aerobic

- It takes place in the presence of oxygen

- Complete breakdown of food/glucose takes place.

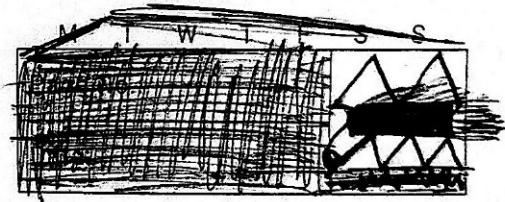
- It produces more energy (38 ATP)

Anaerobic

- It takes place in the absence of O_2 .

- Incomplete breakdown of food takes place.

- It produces less energy (2 ATP)



• End products of aerobic respiration are CO_2 , H_2O & Energy

• In yeast, end product is Ethyl alcohol & Energy, CO_2

~~In muscles the end product of aerobic respiration~~

• In muscles the end product of anaerobic respiration are lactic acid

HIST.

15/12/25

M	T	W	T	F	S	S
Page No.: 5						YOUVA
Date:						

Decline of Gupta Empire

- The later Gupta rulers faced challenges from external invasions.
- The Huna tribe from central Asia repeatedly attacked the empire, weakening the control over North India.
- Rise of powerful Regional rulers led to internal conflict and the dynasty came to an end.

~~The~~

The Kingdoms of South and North-East India

(i) Pallavas in South

- The Pallavas emerged as a powerful dynasty in the south in the present day Tamil Nadu, Karnataka, Andhra Pradesh and Telangana.
- They appeared to have been a tributary power under Satavahanas.

Achievements

- They were great patrons of art and architecture.
- Most of them were devotees of Lord Shiva and are credited with constructing magnificent temples and rock cut caves.

The capital of Pallavas, Kanchipuram was known as the city of the thousand temples, developed as one of the major cities of learning in the South.

(ii) Kamarupa in North Eastern region.

It is ruled by the Varman dynasty extended over the Brahmaputra Valley (Assam) and northern part of present day Bengal and Bangladesh.

An ancient name for Brahmaputra Valley of Assam is Pragjyotsha mentioned in Ramayana and Mahabharata.

The Kamarupa was a prominent cultural and political centre. Temples and monasteries flourished as hubs of learning.

ENGLISH

Q) Prepare a catchy advertisement for a durable and stylish school bag highlighting its features like multiple compartments, water proof materials. Invent necessary details.

Q) You are the manager of 'Paw perfect grooming store'. Draft an attractive newspaper advertisement promoting a new range of pet grooming shampoos, brushes, conditioners, detanglers and paw care items. Highlight special discount, quality assurance and urge pet owners should choose your products.

15/12/25

M T W T F S S
Page No.:
Date: Year:

The Lake Isle of Innisfree

summary

In the poem 'The lake
isle of Innisfree', W.B. Yeats
dreams of escaping the
noises and chaos of urban
life to live peacefully on the
quiet lake isle of Innisfree.
He imagines building a
small cabin, growing beans
and enjoying nature's
gentle sounds. Innisfree
represents tranquility, harmony,
simplicity and spiritual
calmness. Even while standing
in a busy city, the poet hears
the lake water in his heart and
longs to return to that peaceful
place.

Literary devices

- 1) Imagery → "Midnight's all a glimmer, and moon a purple glow."
- 2) Anaphora → "I will arise and go now"
- 3) Alliteration → "I hear lake water lapping with low"
- 4) Symbolism → "The pavements grey" symbolizes the speaker's urgent and unfulfilling urban life.

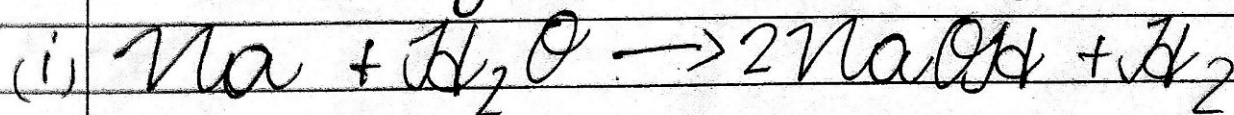
↳ The island symbolizes a desired retreat and return to a simple and a peaceful life.

CHEM.

Type of Metals Based on Reactivity.

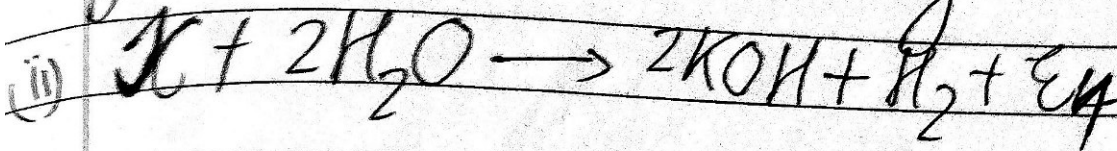
- 1) Most reactive metals
 Eg- Sodium (Na), Potassium (K),
 Lithium (Li) etc.
- 2) Moderate reactive metals-
 Eg- Iron (Fe), Magnesium (Mg),
 Zinc (Zn), Tin (Sn)
- 3) Least reactive / Non
 reactive metals-
 Eg- Gold (Au) & Platinum (Pt)

Reactivity with water.

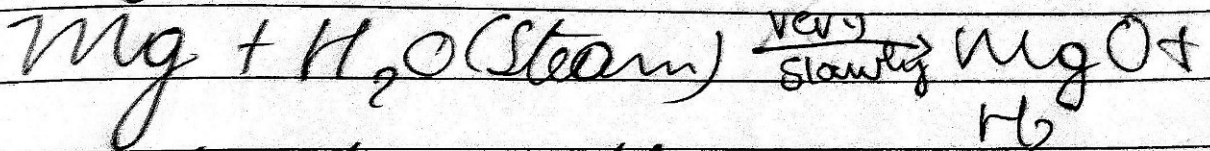
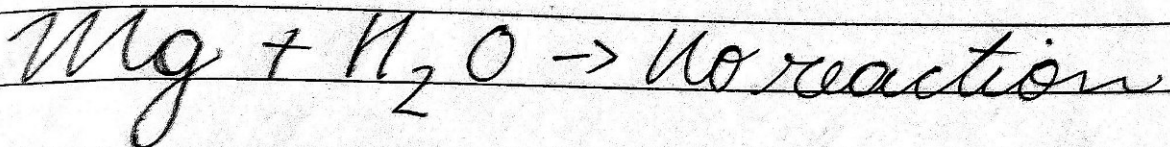


Sodium is highly reactive with water and forms sodium hydroxide and hydrogen gas is released.

During this reaction energy is released w/ hydrogen gas starts burning.



Moderate metals



Reactivity with air or O_2

